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# Best Engineered Bot Criteria

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# Best Engineered Bot Criteria

## Procedure

1. Judges are to visually inspect each team's robot before the competition starts, or have reviewed each robot's Design Process section in the documentation. Ideally, judges will have been able to do both.
    a. Team members are allowed to be present and answer any questions from the judges.
2. Judges will score each robot for the appropriate categories before and during the competition. Judges are to follow the rules and criteria outlined in this document.
3. The highest-scoring robot will be named the winner of Best-Engineered Bot.

## Categories

1. Robots will be judged on their engineering before the competition starts and during the competition.
2. Before the competition starts, robots will be judged on the following:
    **2.1. Material Used**
        2.1.1. The materials chosen for parts used on the robot include all components, with a primary focus on the structural frame and armor.
    **2.2. Structural Integrity**
        2.2.1. The appearance of how parts are designed for strength, including the following:
            2.2.1.1. How the weapon is mounted and secured.
            2.2.1.2. How the armor is mounted and secured.
            2.2.1.3. Are there a number of weight reduction features used that make the robot look less sturdy?
    **2.3. Repairability**
        2.3.1. Does the robot appear to be easily serviceable?
    **2.4. Innovation**
        2.4.1. Does the robot utilize any new and innovative ideas?
3. During their fights, robots will be judged on the following:
    **3.1. Battle Durability**
        3.1.1. How well does the robot continue fighting while damaged in a battle?
        3.1.2. How well does the robot hold up during a battle?
    3.2. **Drive Mobility**
        3.2.1. How well does the robot move around, and how reliable is it?
    **3.3. Weapon Effectiveness**
        3.3.1. How effective is the weapon during battles?
        3.3.2. Does the weapon damage its opponent?
        3.3.3. How reliable is the weapon?

3.3.4. A defensive approach (ex., plow wedge) to control their opponent's movement is to be evaluated if an offensive weapon is not used.

**3.4. Innovation Success**

3.4.1. How successful was the innovation during battles?

# Point System:

1. The following categories have a max allowable of 3 points:
   1.1. Material Used
   1.2. Structural Integrity
   1.3. Robot Repair
   1.4. Battle Durability
2. The following categories have a max allowable of 4 points:
   2.1. Innovative Idea
   2.2. Weapon Effectiveness
   2.3. Drive Mobility
3. The following categories have a max allowable of 5 points:
   3.1. Innovation Success
4. Category Point Examples:
   4.1. Material Used
      4.1.1. The robot uses and is designed with materials that are not subject to holding up in battles (i.e., cardboard, tape, wood).
      4.1.2. The robot uses some standard materials (i.e., a mix of aluminum and wood), but uses some materials that are questionable.
      4.1.3. The robot uses traditional and expected materials (i.e., a combination of metal and plastic).
   4.2. Structural Integrity
      4.2.1. The robot is designed with poor mounting methods or utilizes a number of weight reduction features that impact structural strength.
      4.2.2. The robot is designed with a combination of poor mounting methods or weight reduction features, with strong mounting methods or strong geometry.
      4.2.3. The robot is designed with strong mounting methods and geometry.
   4.3. Robot Repair
      4.3.1. The robot appears not to allow for easy repair serviceability, or requires extensive time to service the robot.
      4.3.2. The robot appears not to allow for easy repairs for some components or systems, but allows others to be serviced easily or promptly.
      4.3.3. The robot appears to be designed with the intention of easy repairs and serviceability.
   4.4. Battle Durability
      4.4.1. The robot is not able to fully function during most of its fights.
      4.4.2. The robot can function properly for about half of its fights.
      4.4.3. The robot can function properly for most of its fights.

4.5. Innovative Idea
    4.5.1. The robot does not have any new or innovative ideas incorporated into its design. All designs are standard and have been done before by other teams.
    4.5.2. The robot incorporates a small innovative idea into its design. A small innovation is classified as a new improvement not related to the weapon or drive system.
    4.5.3. The robot incorporates multiple small innovative ideas into its design, or slight improvements related to the drive and weapon systems.
    4.5.4. The robot incorporates a massive innovative idea into its design. A massive innovation is classified as something new or never seen at a competition before, and would only apply to the weapon or drive systems.
4.6. Weapon Effectiveness
    4.6.1. The robot's weapon or defensive approach is not effective during battles.
    4.6.2. The robot's weapon or defensive approach is only slightly effective during battles.
    4.6.3. The robot's weapon or defensive approach is effective during its battles.
    4.6.4. The robot's weapon or defensive approach is very effective during its battles.
4.7. Drive Mobility
    4.7.1. The robot's drive system mobility is very poor during battles.
    4.7.2. The robot's drive system mobility is poor during more than half of its battles.
    4.7.3. The robot’s drive system mobility is average during most of its battles.
    4.7.4. The robot's drive system mobility is mostly strong and consistent throughout all of its battles.
4.8. Innovation Success
    4.8.1. The robot does not use any innovative ideas, but its design performs strongly in the competition.
    4.8.2. The robot's innovative idea(s) never work properly during its battles.
    4.8.3. The robot's innovative idea(s) have some success but are plagued with issues or do not perform consistently during its battles.
    4.8.4. The robot’s innovative idea(s) have average success but have some issues during its battles.
    4.8.5. The robot's innovation idea(s) perform very well throughout all of its battles.